GECO: GPT-Driven Estimation of 3D Human-Scene Contact in the Wild

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Abstract. Understanding human-scene contact remains a challenging task, as it requires detectors to simultaneously model the contacting body parts, their proximity to scene objects, and the overall scene context. In this work, we introduce GECO, a framework employing Large Language Models (LLMs) with the key insight that language offers a powerful prior to intuitively reason about 3D human-object and human-scene contact based on extensive multimodal world knowledge. By converting a body-vertex formulation to natural language descriptors, we enable zero-shot generation of vertex-level contact directly on the SMPL body. We show that GPT offers a surprisingly competitive baseline close to state-of-the-art detectors on the DAMON dataset. We apply and evaluate different emerging prompting paradigms, highlighting their potential and limitations towards LLM-based human-scene contact estimation.

Keywords: Human-Scene Contact · Large Language Models

1 Introduction

From grasping and locomotion to handling tools and moving heavy items, everyday human activity involves diverse 3D full-body and object contact. With human touch being such a universal aspect in our lives [27], inferring contact from images and videos is attracting ever-increasing attention in applications such as scene understanding, robotics, and mixed reality. However, modeling human-scene and human-object contact is an inherently challenging task, requiring to simultaneously reason about the contacting body parts, their interaction with the various objects, and the surrounding scene context [36]. Existing methods employ a myriad of different neural networks, each aimed at the different particularities of human contact: prior work focuses on self-contact [25], person-person [9] or human-scene [4] contact, while other approaches regress 2D or 3D contact on a body mesh [16,39].

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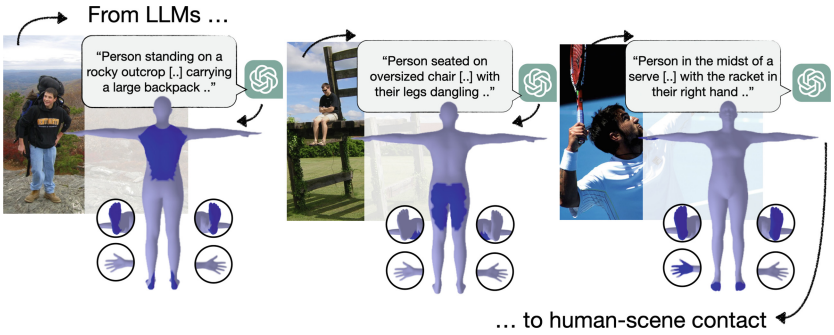

Fig. 1. Our approach leverages foundation models to reason about human-scene contact in an image. We directly infer vertex-level 3D contacts on the human body in a zero-shot, finetuning-free manner (**blue areas**: inferred contact). (Color figure online)

Contrast these special-purpose methods with how humans intuitively perceive contact from an image. For instance, in Fig. 1 (right), one might ask "Do the person's feet touch the ground?" With a quick glance at the image, our cognitive capabilities would rely on "generic world understanding" and contextual cues to infer that the body posture suggest a standing person touching the ground, even when the feet are outside the image frame. Recent advances in multimodal foundation models demonstrate that Large Language Models (LLMs) can leverage such "wealth of world knowledge" [8], serving as strong priors to augment 3D human pose tasks [5,6].

Sharing our intuition, ChatPose [8] employs LLMs to generate 3D body poses from both textual and visual inputs, while ProsePose [34] converts LLM-generated image descriptions into losses for constrained 3D pose optimization. However, these early approaches treat LLM input/outpus as distinct modalities or losses, hence requiring finetuning to the underlying vision task, and they have not fully considered the task of human-object and -scene contact in the wild.

In this work, we introduce GECO, a finetuning-*free* human-scene contact framework based on GPT-4 Vision. Our method relies on converting a body-part formulation to natural language descriptors: when queried with an image, GECO directly generates vertex-level contact in a *zero*/*few*-shot manner. We evaluate our approach on the DAMON dataset [36], showing that GPT provides a surprisingly competitive baseline compared to state-of-the-art models trained on the task. To close the performance gap to finetuned detectors, we investigate emerging LLM prompting paradigms and we show that techniques such as Mixture-of-Experts [29], Self-Consistency [37], and RAG [20] enable better results. We present extensive experiments to highlight the potential and also the current limitations of LLMs towards robust human-scene contact.

2 Related Work

Inferring 3D Contact from Images. Directly inferring contact from images is challenging because it requires understanding the 3D structure of the scene and estimating the contact points. Previous work often breaks the problem down in various subcases, e.g., self-contact [10,25,31], human-human contact [9,26], hand-object contact [1,3,7,40], human-scene contact [14–16,21,30], etc. Here, we focus specifically on cases of human contact with the static scene or objects in the scene. Hassan *et al.* [15] developed a prior for contact events to position people in consistent locations in the scene. Shimada *et al.* [30] infer the points of contact given the 3D scene to guide the reconstruction of the human in the 3D environment. HOT [4] estimates contact between humans and objects by directly returning the pixels that correspond to contact events, without explicit grounding to 3D. BSTRO [16] estimates a more fine-grained output, by indicating the actual locations on the human body (in the form of vertices on the SMPL model [23]) that are in contact with the scene. Since BSTRO is only trained on data from the RICH dataset [16] that has smaller diversity of scenes, DECO [36] annotates in-the-wild images with dense contact information (DAMON dataset) and trains a network that generalizes better on natural images. More recently, LEMON [39] proposed an approach that considers the object of contact in 3D and reasons about the contact points of the object CAD model with the human.

Language Models and Human Pose. Recent work has investigated the use of language for human pose estimation or generation tasks. A popular direction is the use of language for human motion generation [13,28,35,41,42]. Delmas *et al.* [5,6] use natural language to generate or edit 3D human pose examples. ChatPose [8] introduced a multimodal LLM that is finetuned for chatting and producing human poses based on text. Follow-up work, ChatHuman [22], is expanding upon this idea, by incorporating other tools (e.g., for pose estimation, human-object interaction, etc.), that could act as intermediate components for higher-level reasoning tasks. Recently, ProsePose [34] follows a different approach by using descriptions of images provided by LLMs to guide the 3D reconstruction of humans in challenging poses and in human-human interaction scenarios.

3 Methodology

Our guiding principle in this investigation is to explore how foundation models can be used for human-scene contact **without** any additional fine-tuning, loss terms, or modalities. Given an image, our goal is to enable GPT to understand human-object and -scene contact inferred as vertex-level predictions on the SMPL [23] mesh.

3.1 Architecture

Our framework is illustrated in Fig. 2. Given an image $\mathbf{I} \in \mathbb{R}^{H \times W \times 3}$, GECO outputs binary contact predictions on the SMPL [23] body, i.e., $\bar{y}_c \in \{0, 1\}^{6890 \times 1}$.

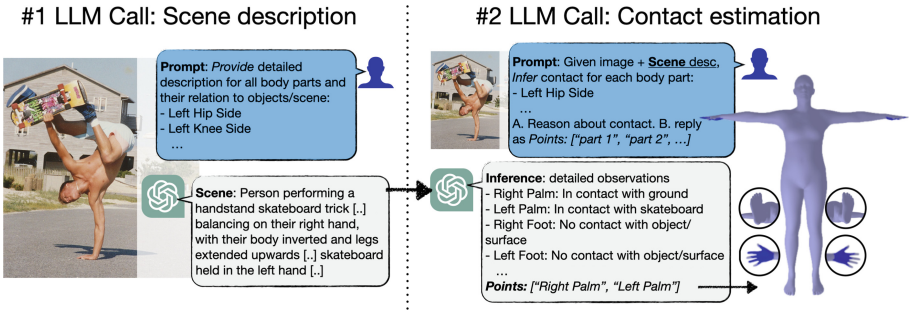

Fig. 2. Method overview. Our approach takes as input an image with a person and follows a two-step ("Socratic agent") paradigm which performs reasoning and inference as two distinct GPT-4 calls. We first obtain a detailed scene description, tasking GPT to particularly focus on SMPL body-parts and their contact in the scene. We then use GPT to predict contact as a list of body-parts, each of which corresponds to a set of body vertices. This list is converted to vertex-level 3D contact predictions on the person's body, hands, and feet (**blue areas**: inferred contact). (Color figure online)

SMPL is a convenient choice for the output of our approach, since its template mesh can be segmented to J body parts, which in turn allows for the part-labeling of contact vertices [36].

Body Mesh Representation. Since we adopt the SMPL body representation, the first step is to define a body part segmentation on the SMPL mesh. Such part-labeling aspect is especially important in the context of LLMs, since it enables us to associate vertices with theit corresponding body parts via their *language* terms. In general there are different ways to segment the SMPL vertices in body parts, e.g., Unite the People [19] defines 6-part and 31-part maps on the SMPL surface. In our case, we want to find a segmentation map that can be easily referred by the language model (i.e., we can refer to each body part with an unambiguous text label) and simultaneously, it is defined at a granularity level that will allow us to precisely localize the contact points. For this reason, we start with an automatic segmentation map that follows the semantic body landmarks defined in [19]. Each vertex is assigned to a body part, based on the body landmark that is closer to it (in terms of geodesic distance). This gives us an initial 87-part segmentation which we further refine manually to aggregate smaller surface patches into larger areas, e.g., combining four different areas for elbow into a single area for each elbow. This leads to the final set of body parts, denoted $\mathcal{P} = \{P_j \mid j = 1, \ldots, J\}$, where $J = 63$ is the number of vertex regions, which are visualized in Fig. 3. Additionally, we obtain a formatted list of the coarser areas, $\mathcal{P} =$ `[‘‘Right Upper Arm’’, ‘‘Right Lower Arm’’, ‘‘Right Hand Palm’’, ...]`. While we also ablate other partitions, we leave further optimization of mesh granularity for future work.

Contact Definition. In effect, the body-vertex partition yields the mapping between each body partition P_j to the corresponding subset of SMPL vertices y_j:

$$P_j \rightarrow \{\{y_j\}_i \mid i \in P_j\}, \quad j = 1, 2, \ldots, 63, \quad i = 1, 2, \ldots, 6890 \tag{1}$$

To obtain contact predictions, we can therefore ask the LMM to generate the set $\bar{P}$ of all body parts that – based on its reasoning – are in contact in the scene. For that, we can obtain the final vertex-level prediction $\bar{y}_c \in \{0, 1\}^{6890 \times 1}$ by setting the corresponding vertices to 1.

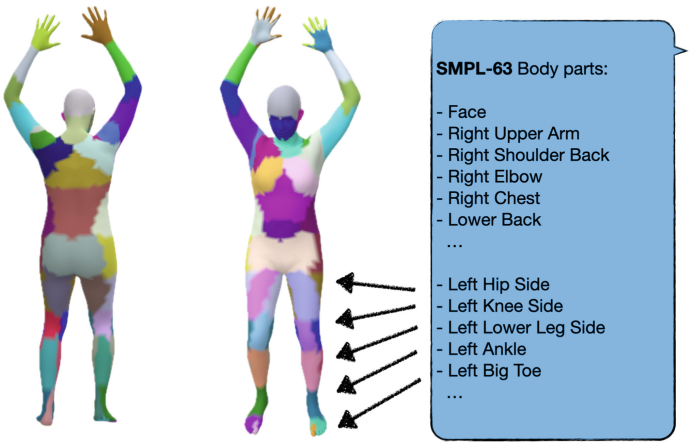

Fig. 3. Color-coded 63 coarse-grained SMPL mesh regions used to ground body-parts to GPT as natural language descriptors.

3.2 Human-Scene Contact Estimation

We follow the two-stage "Socratic agents" paradigm which has been successfully used in other foundation model-based computer vision applications such as embodied AI [24] and scene understanding [33]. That is, we initially prompt the LLM to generate a detailed scene description, and afterwards we ask to infer contact. We find that "reflecting" on the scene separately from inference allows us to efficiently incorporate complex SOTA prompting strategies, e.g., we use the initial description to RAG-search relevant few-shot examples and augment the final prediction.

Reasoning About Human-Scene Contact. In the first stage, we pass the image to GPT-4 and we prompt (`<USER>`) it to generate a comprehensive scene description that covers in detail the overall image context, with particular focus on the person and their interaction with different objects and surfaces. This allows us to leverage zero-shot capability of LLMs over the entire image without the need

to crop a bounding box around the person [8] or prompt with multi-turn dialogues [8]. We show an example of our prompting below. Note how we inject the description of the human-scene contact task directly into the `<SYSTEM>` prompt. We find that this considerably improves how GPT follows instructions – especially by decreasing the likelihood of unresponsiveness (i.e., "I cannot assist you with that request").

Step 1: Prompting GPT-4

`<SYSTEM>` Describe the person's contact with objects and the scene [..]. Follow SMPL and focus on body parts [..]:
- Left Hip Side
- Left Knee Side
- Left Ankle
...
`<IMAGE>` (..image file..)
`<USER>` This is an image of a person interacting with objects and their surroundings. Provide a paragraph with a detailed **scene description**. [..] Enumerate ALL 63 body parts and discuss **in detail** their relation to objects/scene [..].

Contact Inference. In the second stage, we prompt GPT-4 with the image and scene description (Step 1), alongside the definition of the prediction task. As in the previous step, we specify the set of coarse body regions directly in the `<SYSTEM>` prompt. Following chain-of-thought [38] zero-shot (ZS) prompting, we instruct GPT to first write its reasoning before listing the parts in contact. We ask to output the list of all plausible regions by adhering to a specific `<FORMAT>`:

Step 2: Prompting GPT-4

`<SYSTEM>` Infer human-scene contact [..]. Follow SMPL model and consider as possible contact points only [..]:
- Left Hip Side
- Left Knee Side
...
`<IMAGE>` (..image file..)
`<SCENE>` (..scene description from Step 1..)
`<USER>` This is an image and the description capturing a person interacting with objects and their surroundings. Firt, reason which scene aspects suggest contact with body areas [..]. Then, infer the list of all plausible points.
`<FORMAT>` (..e.g., `["Left Foot", "Right Foot", "Right Hand", ...]`..)

3.3 Retrieval-Augmented Contact Estimation

To improve zero-shot performance, we leverage the first stage's scene description and we perform similarity search on train-set images to augment the second-stage inference, akin to Retrieval-Augmented Generation (RAG) in natural language

processing [20]. To this end, we use the DAMON-trainval images and contact ground-truths to create a RAG vectorstore offline by generating a scene description for each image, creating a vectorstore where each entry corresponds to an image. To embed the text-based caption we use FAISS encoder [17] leveraging its LangChain implementation [18].

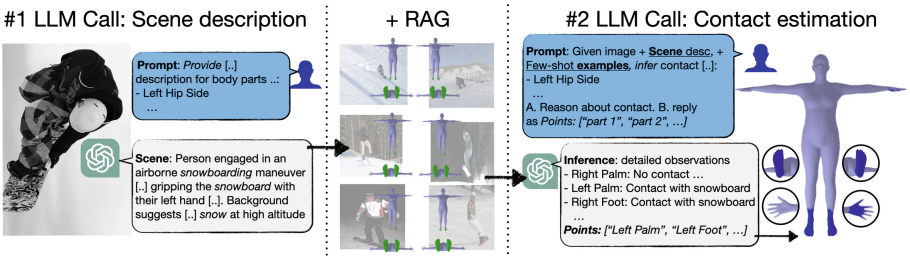

Fig. 4. Retrieval-augmented contact inference. Given an image and its detailed scene description, we RAG-search for relevant train-set images (and their ground-truths) which are then provided as few-shot examples to assist with the prediction at hand (**green**: ground-truth annotations, **blue**: inferred contact). (Color figure online)

We illustrate few-shot (FS) contact prediction in Fig. 4: at runtime, given an "unseen" image (i.e., from DAMON-test that does not appear in the vectorstore), we follow our two-stage inference but before the second stage, we use the generated caption from step 1 to RAG-search for relevant few-shot examples. For instance, consider an image with a person snowboarding; RAG successfully retrieves similar images and their corresponding ground-truths. We find that few-shot leads to a considerable increase in performance (5% F1 score) allowing the few-shot GECO to match the performance of BSTRO [16].

Self-consistency. We consider self-consistency [37] (SC) prompting which has been shown to improve over chain-of-thought. SC involves repeatedly querying the model with the same input image-question pair and employing a consensus mechanism. In the context of GECO binary vertex-level prediction, we use a simple majority-vote scheme over 5 repeated runs per image where, if more than half of the runs agree per contact point, we set the corresponding vertices to 1.

Mixture-of-Experts. Complementing the SC approach, we implement a Mixture-of-Experts (MoE) strategy. Specifically, instead of querying the same model multiple times, we utilize the different models (ZS, FS, SC) with a similar majority voting mechanism to aggregate their binary predictions. We leave other weighted voting schemes for future work.

4 Experiments

Setup. We conduct experiments by deploying 112 GPT-4 Azure OpenAI endpoints (GPT-4 Turbo `vision-preview`) which we query with temperature = 0.1 and "auto" image-detail setting. To reduce the likelihood of unresponsiveness, we select the lowest setting for Azure's GPT content filtering. We leave experimentation with the latest GPT-4o model and other input settings for future work.

Dataset and Metrics. In line with SOTA baselines on human-scene contact, we use the DAMON dataset [36], which includes pairs of (mainly) single-person images with ground-truth contact annotations at the vertex level. As our methodology is finetuning-free, we use the `trainval`-set images only to construct the RAG vectorstore for few-shot at inference time. We evaluate our method on the test split of DAMON, and we follow [16,36] and report the respective count-based evaluation metrics: precision, recall, F1-score, and geodesic error (cm) [16] using DAMON's `metrics()` scripts.

4.1 Results

Following DECO, we compare with the BSTRO, POSA, and DECO baselines on DAMON-test, as shown in Table 1. First, despite its simplicity and without extra prompting schemes, we observe that zero-shot (ZS) GECO is surprisingly close to the baselines in terms of Recall, suggesting that GPT is robust at identifying the contact regions in the image. We observe that both few-shot (FS) and Self-Consistency (SC) prompting improves F1 by 5–6%, bringing the F1 score within 1% from BSTRO.

Table 1. Comparison of DECO with SOTA models on DAMON-test [36].

Methods	Prompting	P ↑	R ↑	F1 ↑	geo. (cm) ↓
BSTRO [16]	–	0.51	0.53	0.46	38.06
POSA [15]	–	0.42	0.34	0.31	33.00
DECO [36]	–	**0.65**	0.57	**0.55**	21.88
GECO	ZS	0.40	0.53	0.40	23.44
	FS	0.40	0.67	0.45	19.26
	SC	0.41	**0.68**	0.46	18.18
GECO + MoE	ZS +FS	0.41	0.59	0.43	16.83
	ZS +FS +SC	0.42	0.62	0.46	**14.58**
	ZS +FS +SC +DECO	0.43	0.68	0.49	11.61

Overall, we note that GPT's main performance comes from to higher recall, based on its reasoning ability at pinpointing body-parts in contact with the scene. However, this comes at a considerable cost of Precision, resulting in worse

F1 scores. This behavior – also observed in POSA – suggests that GPT tends to predict false positives. We postulate that this shortcoming is due to differences in the granularity of (finer) annotations compared to the (coarser) SMPL body-parts. We provide qualitative results to capture such discrepancies.

Furthermore, we observe a 3% improvement in F1 when combining the different ZS +FS +SC schemes into Mixture-of-Experts (MoE) prompting, with MoE matching BSTRO F1 scores. Interestingly, all GPT schemes show strong geo (cm) performance with up to a 7.3 cm improvement in the geodesic error over the closest baseline, DECO. We also ablate augmenting the MoE ensemble with DECO detections, which improves MoE by another 3% in F1 score, as expected.

Table 2. Ablation study for SMPL mesh and its granularity design choice.

Methods	Body Mesh	P ↑	R ↑	F1 ↑	geo. (cm) ↓
GECO - ZS	SMPL 87-parts	0.39	0.51	0.39	26.30
	SMPL 63-parts	0.40	0.53	0.40	23.44

Finally, we ablate the effect of dividing the mesh surface of the SMPL model in body parts in different levels of granularity. First, we experiment with our initial 87-part segmentation map that corresponds to the SMPL 87-parts baseline of Table 2. Using this body part segmentation leads to reasonable results, but we noticed that the output included mistakes that could be attributed to the very fine-grained nature of this map (e.g., differentiating between "left" and "right" lower back. After a manual aggregation of some fine-grained areas into larger patches, we obtain the SMPL 63-parts version used in Table 2. We observed consistent improvements and less confusion when referring to the different body part in the results with this version, so this is what we adopted as the main version for our experiments.

4.2 Qualitative Results

Contact Estimation Granularity. Figure 5 shows examples where the granularity of the selected SMPL body regions inherently cannot reflect the (extremely) finer granularity of the corresponding labels. GPT correctly captures the overall semantics of each scene – e.g., GPT-4's caption (bottom image) reads "person with their legs crossed, with the left [..] on top of the right knee" – showing good recall with respect to the ground-truth. However, the coarse nature of the surface body segments would result in a large portion of body-vertices being false-positives, hence hurting overall the Precision of GECO.

Left-Right GPT Errors. Prior work reports GPT's struggles with reliably and correctly referencing left and right limbs [34]. Indeed, we encountered inconsistency issues, as shown in Fig. 6, for both hands and legs. Since – unlike [34] – we do not treat left-right pairs as coarse groups, we find that in practice better

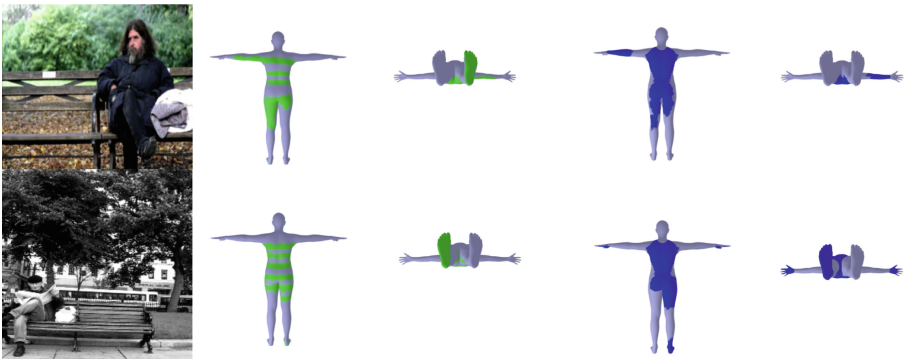

Fig. 5. Fine ground-truth annotations *vs.* coarse body-regions. While GECO achieves high Recall because GPT captures the overall image semantics, Precision suffers due to several vertices being considered false-positives. For each example, we show the input image, the ground-truth annotations (**green**) and inferred contact from GECO (**blue**). (Color figure online)

prompting, especially multiple SC runs, help better disambiguate the chirality. It is worth mentioning that, in our ongoing preliminary trials with the latest GPT-4 (Omni) version, we find "left-right" errors to be reduced considerably. We leave GPT-4o integration as future work when available at a large scale – current Azure Requests-per-Minute constraints prevent wider consideration.

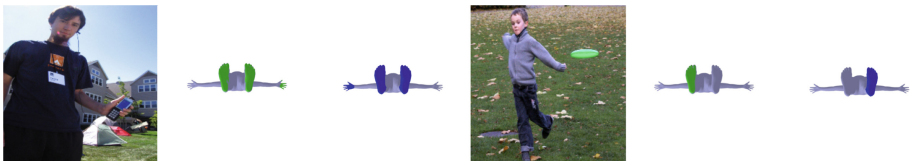

Fig. 6. Confusing left-right body parts. One common issue of GPT is that it correctly identifies the contact events (i.e., the hand is in contact), but attributes it to the wrong body side. For each example, we provide the input image, the DAMON annotations (**green**) and the inferred contact from GECO (**blue**). (Color figure online)

Front-Back GPT Errors. Besides chirality, an interesting source of error – which disproportionally impacts metrics due to the extent of the body-parts involved – is people lying face down (Fig 7). Surprisingly, GPT's chain-of-thought reasoning collapses in those cases, with the drawn conclusion *not* matching the initial (correct) reasoning. For instance, the scene description for Fig. 7 (top) reads "person on bed positioned on their stomach [..] reading a book [..] with their arms bent at the elbows." However, in the final inference, GPT overrides the reasoning for

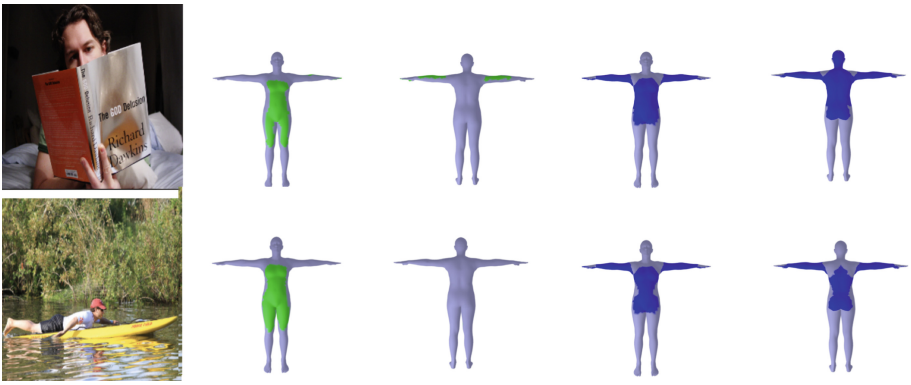

Fig. 7. Confusing front-back body parts. Sometimes, GPT identifies that there is a contact with a particular body part (i.e., with the torso), but fails to return the correct side (front-back) that is in contact. For each example, we provide the input image, the DAMON annotations (**green**) and the inferred contact from GECO (**blue**). (Color figure online)

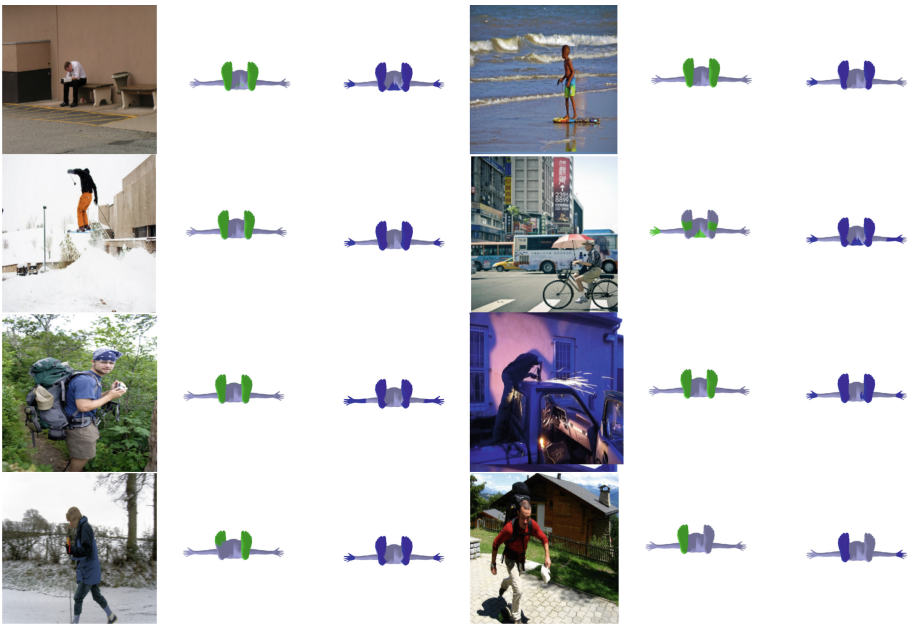

Fig. 8. "Oracle-error" analysis. We find that GPT reliably captures object grasping, as it can infer palm/hands contact from "person holding object X" observations with high confidence. Interestingly for high-confident errors, we note sporadic issues with the corresponding ground-truth. For each example we provide the input image, the DAMON annotations (**green**) and the inferred contact from GECO (**blue**). (Color figure online)

all regions on the back of the body as "*Likely* in contact with the pillow or bed with person lying on bed." Notice the word 'Likely' here: this behavior resembles strong "Blind agent" tendencies also observed in embodied AI foundation models [24], where the LLM would blindly trust its "language/common sense prior" (i.e., "lying in bed" so "likely" on their back) rather than the observation per se. We find that behavior to persist even across multiple SC runs.

4.3 "Oracle-Error" Analysis

Besides our qualitative analysis, we conducted further experiments to systematically understand failure modes of the model. Following *oracle-error* schemes from other computer vision tasks [2] to identify sources of error, we focus on instances where GECO exhibits strong prediction confidence yet fails to match the ground-truth. As our formulation does not provide explicit confidence scores, we repurpose our SC code to repeat the same inference 10 times per image, and we identify as instances with high confidence the ones where the results match in 8/10 repetitions.

One interesting emerging aspect, shown in Fig. 8, relates to cases where GPT (properly) identifies people grasping objects which are not labeled in DAMON. Overall, it is intuitively expected that GPT can reliably correlate higher-level textual observations such as "person holding a camera" or "person cross-country skiing [..] holding ski poles for propulsion" with hand/palm contact. While outside the scope of our work, we motivate that such aspect could be sufficiently leveraged in lieu of expensive contact annotations, as also discussed in recent 3D pose LLM works [34].

4.4 Limitations and Future Work

While GECO provides a robust initial approach to GPT-driven contact estimation, it has limitations which offer several interesting directions to improve as future work. First, it is still likely that GPT would refuse answering some queries, e.g., COCO images depicting persons holding a knife. As a rudimentary solution, we repurpose the SC scripts to retry the same failed request multiple times with higher temperatures > 0.1 until getting an answer. While partially effective, such brute force solution increases cloud cost. On that note, RAG- and ensemble-based prompting are also inherently expensive, so we note that GECO might not be scalable to larger datasets. We motivate the need for optimizations on the LLM-system stack, such as token reduction [12,32] or request batching [11,33].

Second, in terms of modeling, the granularity of the SMPL mesh could be further improved to better reflect the "language bias" on how LLMs perceive the "particularities" [34] of human contact. For instance, we mentioned the potential room to improve by resolving left-right and front-back confusions. We aim to address these aspects in future work. Last, we note the tremendous potential of leveraging foundation models to unlock multiclass human-object contact

inference. To this end, as DAMON [36] provides object-categories, it would be interesting to revise our approach beyond binary predictions.

5 Conclusion

In this work, we present GECO, a finetuning-free GPT-based framework that leverages foundation model's "world understanding" to intuitively reason about 3D human-scene contact. Our key intuition is to enable *zero/few*-shot contact prediction by associating body-vertex regions via natural language descriptors, without the need for the expensive annotations and detector training. At runtime, GPT generates the list of body-areas in contact which we convert to vertex-level predictions directly on the SMPL mesh. Experiments with SOTA prompting schemes show that GPT offers a competitive baselines compared to prior work. We are excited to see this rich, emerging space at the vibrant intersection between digital humans and foundation models which will open up new ways to reason about 3D human modeling.

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